

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1083

Southgate 138 kV
December 03, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1083. The Interconnection Customer proposes a 490.1 MW natural gas combined cycle facility interconnecting at the Southgate 138 kV substation in the New Jersey area. The request was received on October 20, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 274 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,307,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Southgate / Kelly Lake 138kV Line	Oyster Creek / Southgate Substation 138kV	185.2	256	100.0
Southgate / Oyster Creek 138kV	Kelly Lake / Southgate Substation 138kV L	177.0	267	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Southgate / Kelly Lake 138kV Line	Rebuild line with higher-capacity conductor	\$3,999,000	100% (\$3,999,000)
Southgate / Oyster Creek 138kV	Rebuild line with higher-capacity conductor	\$3,308,000	100% (\$3,308,000)
	Total allocated to project		\$7,307,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.